

SmartCity AR

Re-imagining Cape Town's Foreshore Freeway

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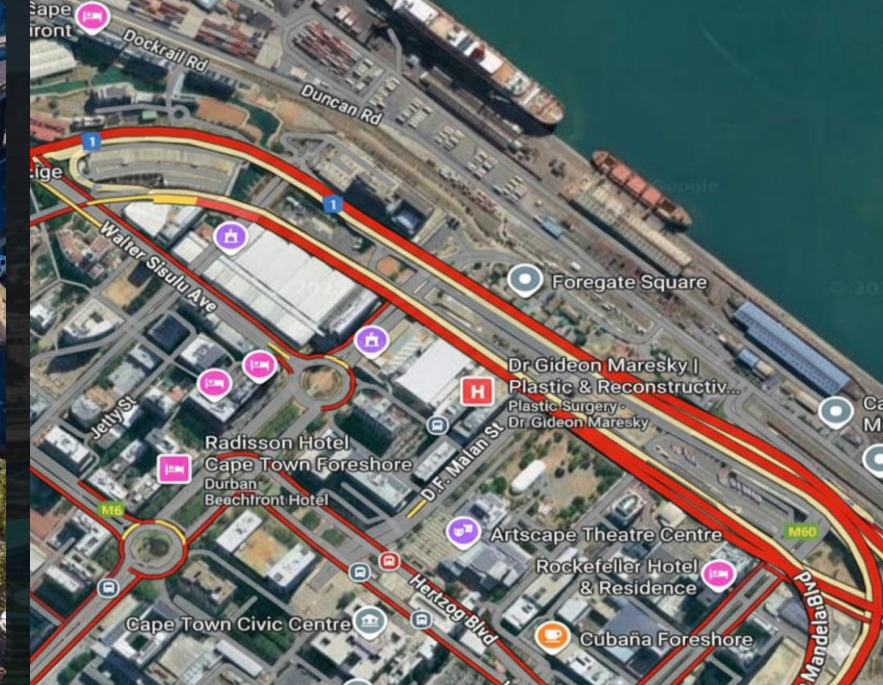
Transforming Cape Town's Foreshore Freeway Precinct with AR

Project Focus:

- Cape Town's Foreshore Freeway Precinct
- Traditional maps fail to show dynamic impact & public participation is limited.
- Frustration sitting in traffic – it feels unproductive and wasted time



Unfinished Bridge



Traffic Congestion



Chronic Congestion

Peak-hour traffic overwhelms existing infrastructure



Underutilized Land

Valuable urban space remains disconnected and underdeveloped



Visualisation Gap

2D tools cannot effectively communicate dynamic urban transformations

The SmartCity AR Solution

- This innovative approach enhances public analysis, facilitates informed decision-making, and ensures transparent stakeholder engagement throughout the redevelopment process.
- Overlaying digital models onto physical spaces, AR transforms abstract concepts into tangible, comprehensible visions.



SmartCity AR Application

- Augmented reality technology bridges the visualisation gap, enabling planners, officials, and citizens to experience proposed changes in real-time, three-dimensional context.

Requirements: Defining the Solution

Functional Must-Haves:

- Scenario Toggle: Sustainable vs. Unsustainable.
- Layer Toggle: Roads, Water, Electricity, etc.
- Traffic Simulation & Camera Navigation.

Non-Functional Targets:

- Performance: Load $\leq 5s$ & Maintain ≥ 30 FPS.
- Compatibility: Mid-Range Android Devices (ARCore).



Sustainable scenario



Unsustainable scenario

Architectural Design: Modular & Maintainable

Scene Manager

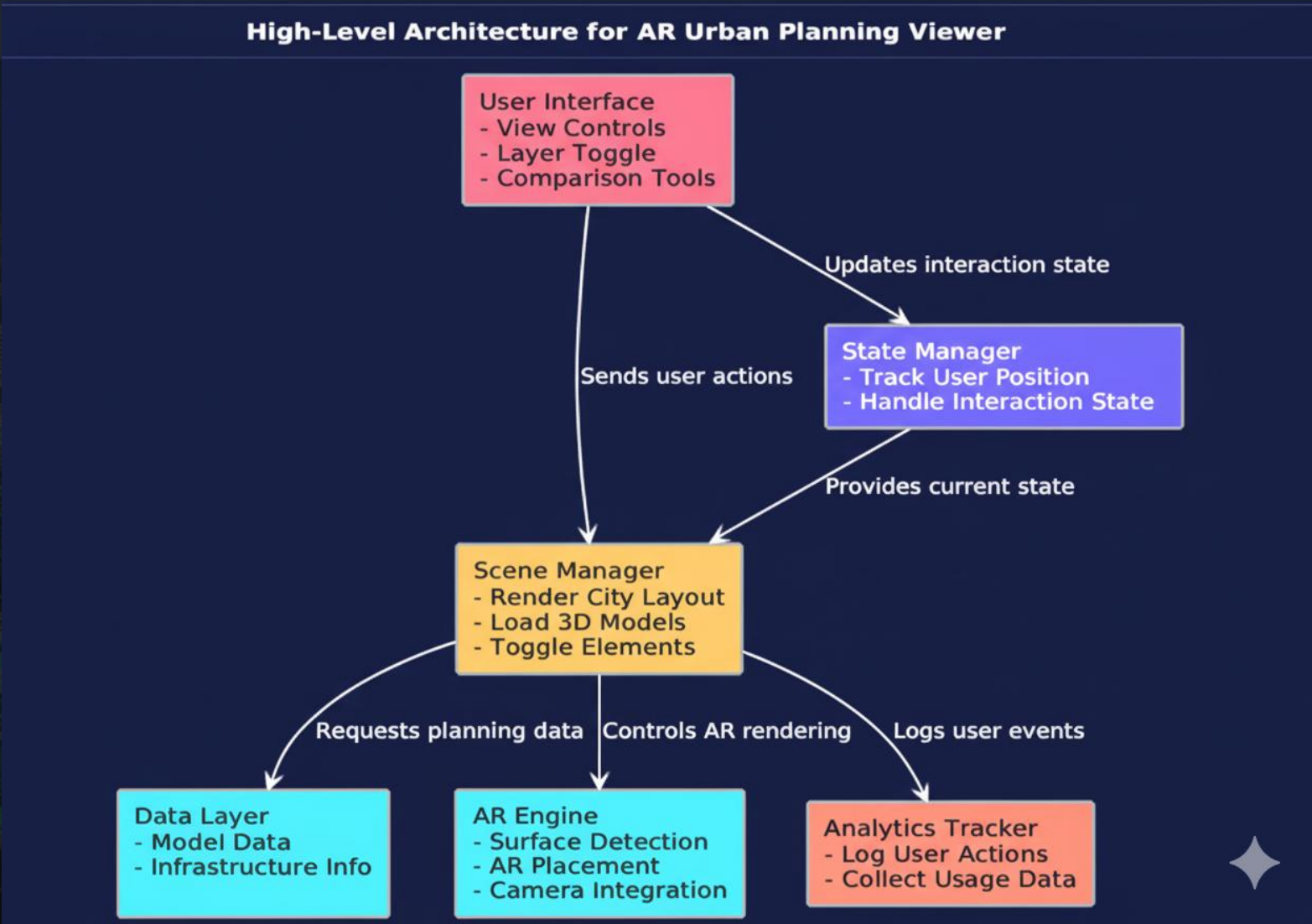
Handles 3D Models/Layers

AR Engine

ARCore/Surface Detection

Controller

Input/State Management



□ Design Logic: Separation of UI, Data, and Rendering.

Implementation & Key Components

Core Stack



Engine:

Unity 2022.3 (C#).



AR Framework:

AR Foundation/ARCore.



Modeling:

Blender + Blender GIS Add-on.



Key Functionality Developed

TrafficSpawner (Object Pooling)



CameraMoveTrigger (Smooth Interpolation)

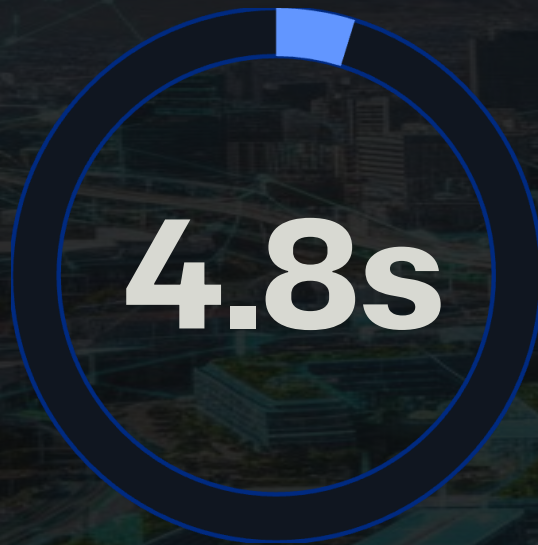


Testing & Project Validation

Testing Strategies:

- Functional: (Toggles, Traffic, Scene Reset).
- Performance: (NFRs $\leq$ 5s load, $\geq$ 30 FPS).

Key Performance Results:



Scene Load

Average 4.8s (Target $\leq$ 5s) - PASS



Frame Rate

Average 30-46 FPS (Target $\geq$ 30 FPS) - PASS



Rendering Latency

$\leq$ 90ms (Target $\leq$ 100ms) - PASS

Future Work:

- Known Issue: Performance dip under stress.
- Recommendation: Implement Level-of-Detail (LOD) optimisation for traffic.

SmartCity AR

By Mohamed Asad Bandarkar

References & Conclusion

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Thank you. Application demonstration video next.